

Voice-Controlled Home Automation System Using ESP32 and Google Assistant

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1 Abstract

This project presents a Voice-Controlled Home Automation System using ESP32, Google Assistant, and Sinric Pro. The system enables users to control connected devices through voice commands. Google Assistant processes user commands and communicates with Sinric Pro Cloud, which forwards the commands to the ESP32 over Wi-Fi. Three LEDs are used to represent smart appliances and demonstrate real-time voice-controlled operation. The system provides a low-cost and scalable solution for smart home automation applications.

2 Introduction

The Internet of Things (IoT) has significantly transformed the way devices communicate and interact. Smart home automation systems have become increasingly popular due to their convenience, efficiency, and remote accessibility.

Voice-controlled automation further enhances user experience by allowing hands-free interaction with devices. In this project, Google Assistant is integrated with ESP32 using Sinric Pro Cloud to control devices through voice commands. The implementation demonstrates a practical approach to modern smart home automation using affordable hardware and cloud-based communication.

3 Objectives

- To develop a voice-controlled home automation system using ESP32.
- To integrate Google Assistant with ESP32 using Sinric Pro.
- To control multiple devices using voice commands.
- To demonstrate IoT-based wireless communication.
- To provide a scalable smart home automation solution.

4 Components Used

- ESP32 Development Board
- Red LED
- Green LED
- Blue LED

- Breadboard
- Jumper Wires
- Wi-Fi Network
- Google Assistant
- Sinric Pro Platform
- Arduino IDE

5 System Architecture

The system architecture consists of four main components: Google Assistant, Sinric Pro Cloud, Wi-Fi Network, and ESP32. The user issues voice commands through Google Assistant. The command is transmitted to Sinric Pro Cloud, which processes and forwards it to the ESP32. Based on the received command, the ESP32 controls the corresponding output pin connected to the LED.

6 Circuit Diagram

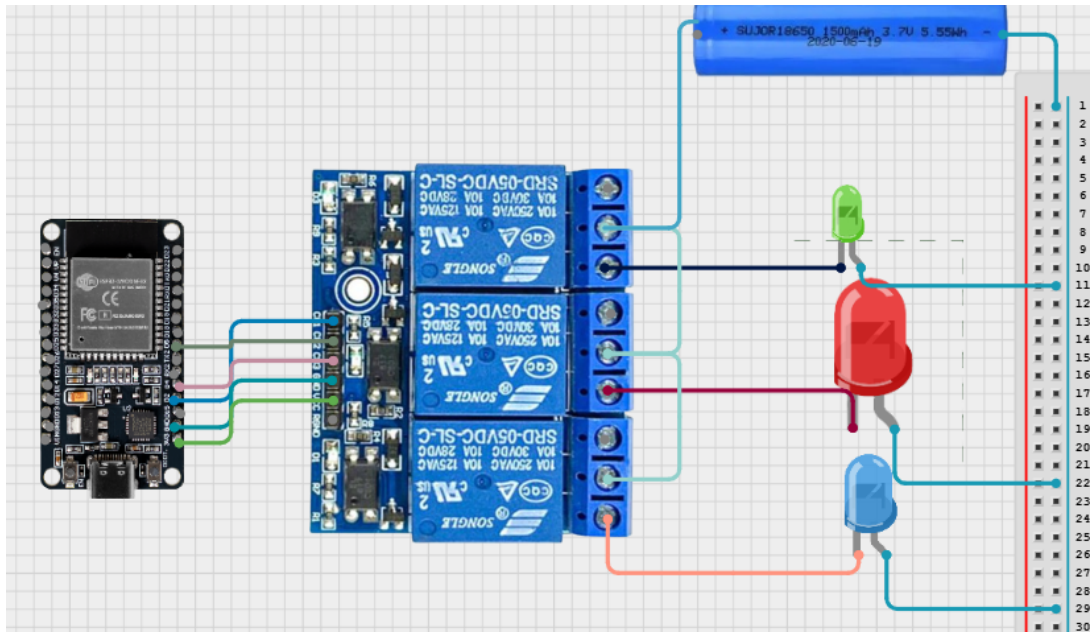


Figure 1: Circuit Diagram of Voice-Controlled Home Automation System

The hardware circuit consists of an ESP32 development board connected to three LEDs through GPIO pins. GPIO4 controls the Red LED, GPIO15 controls the Green LED,

and GPIO2 controls the Blue LED. Each LED is connected through a current-limiting resistor to ensure safe operation. The circuit demonstrates the control of multiple devices using voice commands.

7 Methodology

The implementation begins with configuring the ESP32 and connecting it to a Wi-Fi network. Sinric Pro credentials are used to authenticate the device with the cloud platform. Three virtual devices are created and linked with Google Assistant.

When the user issues a voice command, Google Assistant forwards the request to Sinric Pro Cloud. The cloud service communicates with the ESP32 and updates the state of the corresponding GPIO pin. The selected LED then turns ON or OFF according to the command received.

8 Working Principle

The ESP32 continuously maintains communication with Sinric Pro Cloud through Wi-Fi. Voice commands are captured by Google Assistant and processed in the cloud. Once the command is validated, Sinric Pro sends the instruction to the ESP32. The microcontroller identifies the target device and changes the state of the corresponding output pin.

The Red LED is connected to GPIO4, the Green LED to GPIO15, and the Blue LED to GPIO2. Each LED responds independently to voice commands, demonstrating multi-device control capability.

9 Software Implementation

The project was implemented using Arduino IDE and the Sinric Pro library. ESP32 was programmed to connect to Wi-Fi and communicate with Sinric Pro Cloud. Three virtual devices were configured and mapped to GPIO4, GPIO15, and GPIO2 respectively.

The software continuously listens for incoming commands from Sinric Pro and updates the output states in real time. This ensures reliable and responsive device control.

10 Testing and Validation

The system was tested using multiple voice commands through Google Assistant. Commands for turning ON and OFF individual LEDs were executed successfully. The communication between Google Assistant, Sinric Pro, and ESP32 remained stable throughout testing.

The response time was observed to be minimal, and all devices responded correctly to user commands. This confirms the reliability of the proposed system.

11 Results

The developed system successfully established communication between Google Assistant, Sinric Pro Cloud, and ESP32. Voice commands were accurately processed and executed in real time.

The Red, Green, and Blue LEDs responded correctly to user commands. The system demonstrated stable performance, low latency, and reliable operation during testing.

12 Advantages

- Hands-free operation using voice commands.
- Real-time wireless control.
- Easy integration with Google Assistant.
- Low-cost implementation.
- Scalable architecture.
- User-friendly operation.

13 Applications

- Smart Home Automation
- Smart Lighting Systems
- Voice-Controlled Appliances
- Office Automation
- Assistive Technology
- IoT-Based Device Control

14 Future Scope

Future improvements may include relay-based control of AC appliances, mobile application integration, energy monitoring, cloud-based data analytics, AI-driven automation, and support for additional smart home devices.

15 Project Demonstration Video

The complete working demonstration of the project is available at:

https://drive.google.com/file/d/1QVRz4gogtG4Z4sCtLEmQrsCyxuU_RzDf/view?usp=drive_link

16 Conclusion

This project successfully demonstrates a Voice-Controlled Home Automation System using ESP32, Google Assistant, and Sinric Pro. The implementation provides a practical and efficient solution for controlling devices through voice commands. The system is reliable, scalable, and suitable for modern IoT-based smart home applications.

17 References

1. ESP32 Technical Documentation
2. Sinric Pro Documentation
3. Google Assistant Documentation
4. Arduino IDE Documentation
5. Internet of Things Fundamentals